

Development of a Mathematical Model for the Simulation of Ex Vivo Ocular Tissue Clearing

Proof of “Refractive Index From Changes in CUBIC Concentration” Equation

Proof. Using the assumption that the refractive index is proportional to the volume that CUBIC occupies within ocular tissue matrices, the following equation can be created to model the refractive index of each tissue layer:

$$I_{\text{layer}} = I_0 \left(\frac{C_{\text{cubic}}}{\varepsilon V_{\text{layer}}} \right).$$

In this equation, I_0 is equivalent to 1.52, or the refractive index of CUBIC-R+, which is the CUBIC mixture used for refractive index matching.